Information Security
Assignment 1
Section: ES

Muhammad Imran Butt
L1F24BSCS0546

Problem 1 :-
Vigenére Cipher

a) P = DEFEND THE EAST WALL

K = KEY

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S

19	20	21	22	23	24	25
T	U	V	W	X	Y	Z

Key: KEYKEYKEYKEYKEYKE

	D	E	F	E	N	D	T	H	E	E	A	S	T	W	A	L	L	
	3	4	5	4	13	3	19	7	4	4	0	18	19	22	0	11	11	
+	K	E	Y	K	E	Y	K	E	Y	K	E	Y	K	E	Y	K	E	
=	10	4	24	10	4	24	10	4	24	10	4	24	10	4	24	10	4	mod 26
C:	N	I	D	O	R	B	D	L	C	O	E	O	D	A	Y	V	P	
	13	8	3	14	17	1	3	11	2	14	4	12	3	0	24	21	15	

Encrypted:

B) Ciphertext: GIKLFO ZHA ICD SO YHIV

Icey: ICEY ICEY ICEY ICEY ICEY ICE

	6	8	10	11	5	14	25	7	0	10	3	18	14	25	7	8	21	
	G	I	K	L	F	O	Z	H	A	I	C	D	S	O	Y	H	I	V
	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
P =	I	C	E	Y	K	E	Y	I	C	E	Y	I	C	E	Y	K	E	Y	mod 26
	10	4	24	10	4	24	10	4	24	10	4	24	10	4	24	10	4	

P = WEMBBQPDCAZUEVJYR

Problem 02
Play Fair

A)

P = ATTACK AT DAWN

K = MONARCHY

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Pairs

AT TA CK AT DA WN

Encrypted

~~BRSQE~~

RS SR DE RS

BR NX

Dycrypted

AT TA CK AT DA WN

B) P = MEET ME AT 3PM

key = security

S	E	C	U	R	I
T	Y	A	B	D	F
G	H	J	K	L	M
N	O	P	Q	V	W
X	Z	0	1	2	3
4	5	6	7	8	9

Pairs :- ME ET ME AT 3P MX

Encrypt :-

HI SY HI BY EW GN 3

Decrypted

ME ET ME AT 3P MX

CamScanner